

A SCALABLE BENCHMARKING TOOLKIT FOR AI-DRIVEN IMAGE SEARCH

Sage Image Search



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ABSTRACT

Evaluating AI systems at scale, particularly domain-specific tasks, remains difficult due to the lack of extensible and reproducible benchmarking frameworks. We present a benchmarking toolkit for large-scale image search, designed to enable systematic evaluation, comparison, and improvement of retrieval systems across models, datasets, and deployment environments. Our toolkit supports the full benchmarking lifecycle: dataset ingestion, query generation, relevance annotation, and metric-driven evaluation. It enables the construction of domain-specific benchmarks with rich metadata and controllable difficulty, supporting use cases such as environmental monitoring and wildfire detection. The framework integrates standard retrieval metrics (e.g., NDCG, precision, recall) alongside diversity-aware measures such as intra-list similarity, providing a more complete view of system performance. A key feature is its tight coupling with real-world retrieval pipelines. The toolkit evaluates search systems that combine semantic embeddings, keyword retrieval (BM25), and reranking models, while remaining model-agnostic and infrastructure-independent. This allows researchers to benchmark different architectures, indexing strategies, and ranking approaches under consistent conditions. The system is designed with reproducibility and scalability as first-class concerns: experiments are fully scriptable, benchmarks are portable, and evaluation workflows can be applied to both historical datasets and live data streams. When deployed on large-scale datasets such as those from the Sage Continuum (tens of millions of images), the toolkit enables rigorous, system-level analysis of trade-offs between retrieval quality and latency. This work establishes a general-purpose, extensible foundation for benchmarking multimodal retrieval systems in domain-specific settings.



view blog post:

MOTIVATION

Large-scale, multimodal datasets such as the tens of millions of images collected by the Sage Continuum require more effective retrieval than traditional metadata-based search can provide. While AI-powered image retrieval systems have advanced rapidly, there remains a fundamental gap in how they are evaluated, especially for domain-specific tasks where general benchmarks fail to capture real-world complexity. As a result, it is difficult to compare models, understand quality-latency trade-offs, or measure true system improvements. **This work addresses the evaluation gap by enabling systematic, reproducible evaluation under domain-specific, production-like conditions.**

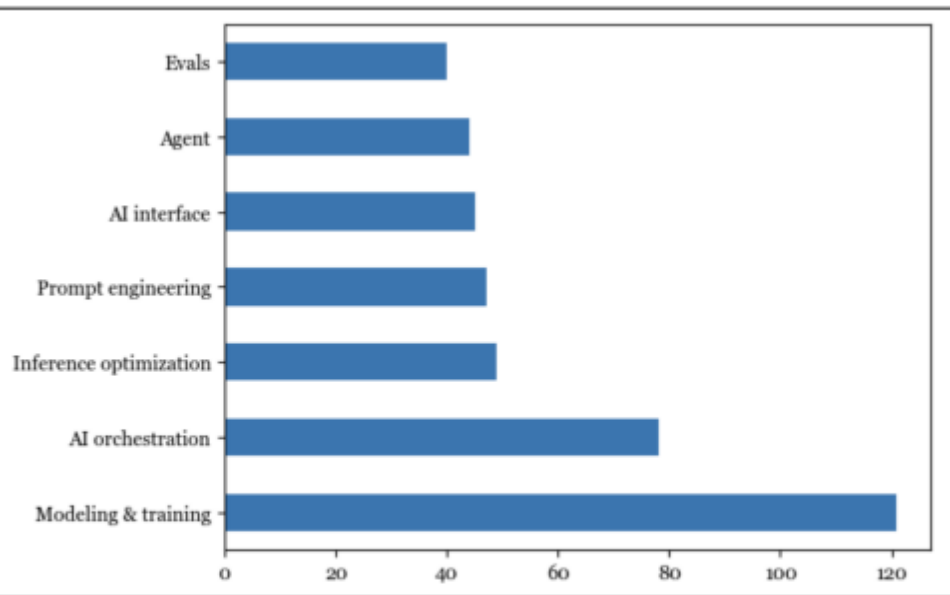


Figure 3-3. According to data sourced from my list of the 1,000 most popular AI repositories on GitHub, evaluation lags behind other aspects of AI engineering in terms of open source tools.

Figure sourced from: Chip Huyen, AI Engineering, O'Reilly Media, 2024. ISBN: 978-1-098-16630-4.

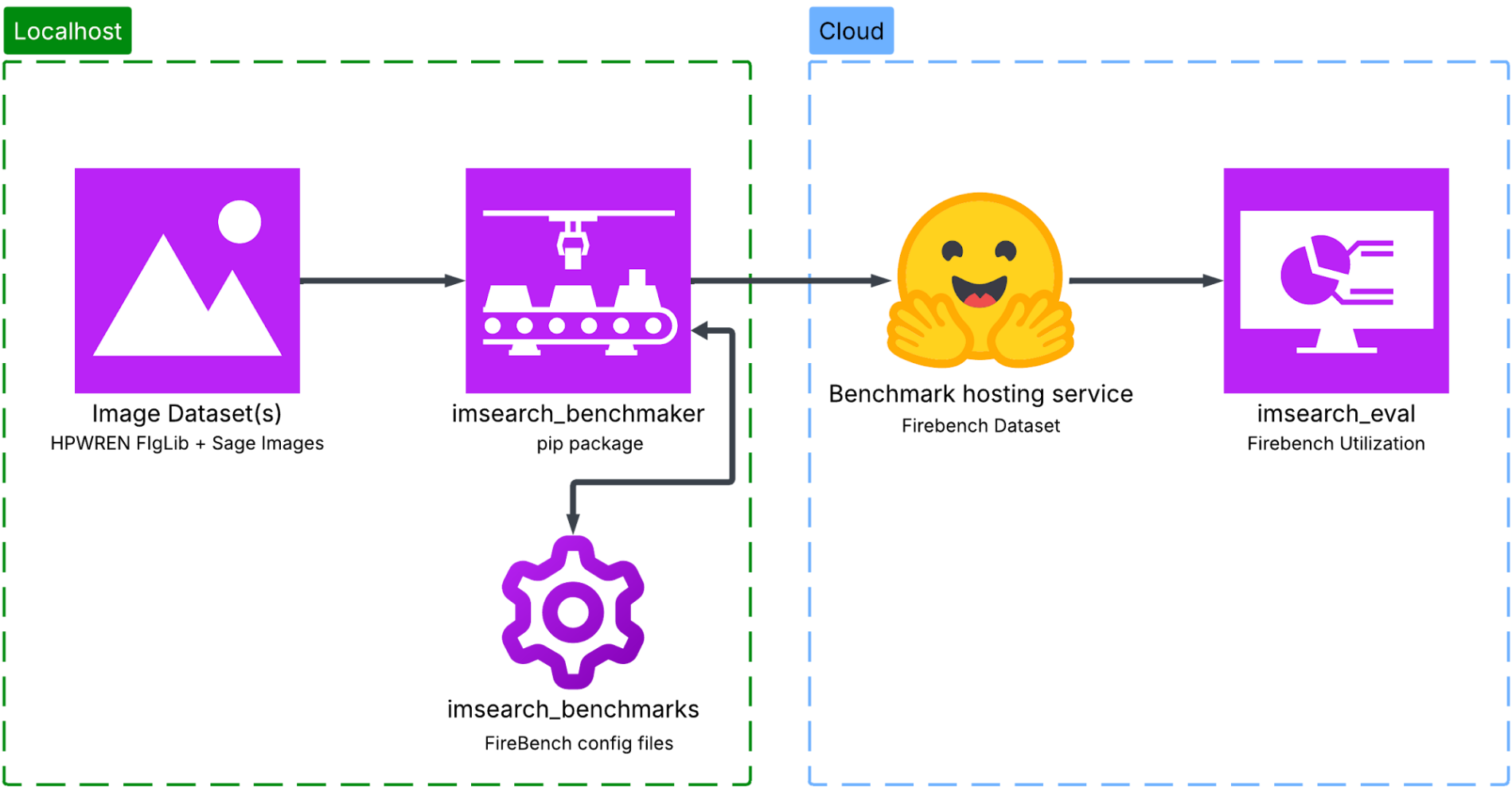
METHODS

- Benchmark Generation:** Domain-specific datasets with controlled query difficulty, rich metadata, and relevance annotations.
- Evaluation:** Benchmarks executed on the live system with metric-driven analysis.
- Scalable Execution:** Parallel benchmarking on HPC enables comparison across models, prompts, and configurations.
- System-Level Analysis:** Joint monitoring of retrieval quality and latency under realistic, production-like conditions.

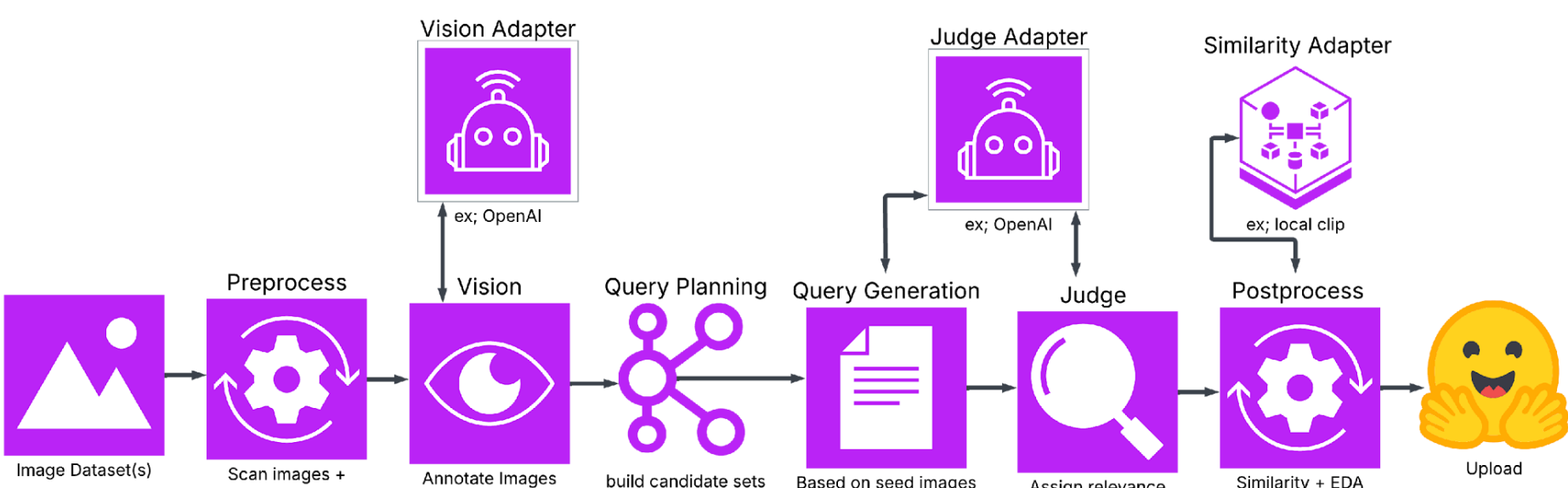
ECOSYSTEM

- imsearch_benchmark** - benchmark dataset generation
- imsearch_benchmarks** – curated domain benchmarks generated by benchmarker
- imsearch_eval** – evaluation framework
- Grafana Dashboard** – live latency monitoring

Toolkit Workflow



Benchmark Maker Workflow (imsearch_benchmark)



View imsearch_benchmark Repo:

RESULTS

Example toolkit outputs comparing AI system versions with domain-specific breakdowns (e.g., fire science). Shown: Hit Rate and NDCG; additional metrics and domain-specific taxonomies are also available.

- Hit Rate:** Percentage of queries that had at least one relevant image returned in the top k images.
- Normalized Discounted Cumulative Gain (NDCG):** Measures how well a system ranks relevant results near the top compared to an ideal ranking (range 0-1).



View Evaluation Example:

WHY AI AS A JUDGE?

- Scalable:** Fast, easy to deploy, and far cheaper than human evaluation
- Reasonably Reliable:** Studies show high agreement with human evaluators
 - GPT-4 reached 85% AI-human agreement vs. 81% human-human agreement (Zheng et al., 2023)
- Insightful:** Can provide explanations for its decisions
- Key point: “Even when AI judgements aren’t as good as human judgements, they might still be good enough to guide an application’s development and provide sufficient confidence to get a project off the ground”

LIVE BENCHMARKS

Each benchmark includes a config.toml, tools, readme, exploratory data analysis, and dataset

Dataset	Domain
FireBench	Fire Science 🔥
CommonObjectsBench	General Objects & Scenes 🌍
CloudBench	Nephology ☁️
SageBench	Sage Metadata 🌲
Inquire	Biology 🌿



View Benchmarks:

NEXT STEPS

- Expand** benchmark coverage to **new domains** relevant to Sage users
- Generalize** the toolkit to support **new tasks** beyond image search
- Enhance packaging and usability** to streamline adoption of the AI benchmarking toolkit

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